

1 Introduction

The Cosmic Microwave Background (CMB) provides a unique window to study the early stages of the Universe and its structure formation. In particular, measurements of CMB polarization provide a powerful probe of primordial gravitational waves generated during the inflationary epoch. These gravitational waves imprint a distinctive pattern in the CMB polarization, known as B-modes.

Detecting primordial the B-mode would constitute direct evidence for inflationary gravitational waves, enabling a measurement of the tensor-to-scalar ratio, and thereby constraining the energy scale of inflation. Such a discovery would open an unprecedented observational window onto high-energy physics, far beyond the reach of terrestrial experiments.

However, its detection does not depend solely on instrumental sensitivity. A successful discovery critically relies on accurate component separation, since the cosmological signal is contaminated by polarized Galactic emission with significantly higher amplitude.

1.1 Galactic foregrounds and data

In this figure, you can see the frequency dependence of the CMB in brightness temperature.

2 Methodology

2.1 Angular power spectra (APS)

2.2 Noise estimation

2.3 Window functions and CMB subtraction

2.4 Error estimation

Foreground model

2.5 Synchrotron emission

2.6 Thermal dust emission

2.7 Synchrotron-dust spatial correlation

2.8 Bayesian inferenced and MCMC

2.9 Masks and summary

3 Results

3.1 Synchrotron amplitude ratio

3.2 Dust amplitude ratio

3.3 Synchrotron spectral index

3.4 Dust spectral index

3.5 Synchrotron-dust correlation coefficient

3.6 Synchrotron slope

3.7 Dust slope

4 Conclusions